American Computer Science League

2020-2021 ● Contest 4: Shorts ● Senior Division

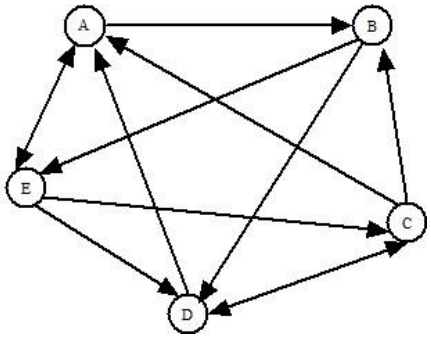
1. Graph Theory

Given the set of vertices {A, B, C, D} and the set of directed edges {AB, AA, AC, AD, BA, BC, CC, DC}, how many paths are there of length 3?

A. 21
B. 22
C. 25
D. 26
E. 28

2. Graph Theory

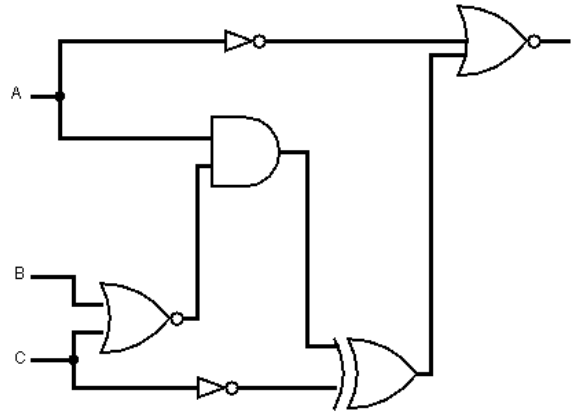
How many cycles are there in the following directed graph?



A. 15
B. 16
C. 17
D. 18
E. 19

3. Digital Electronics

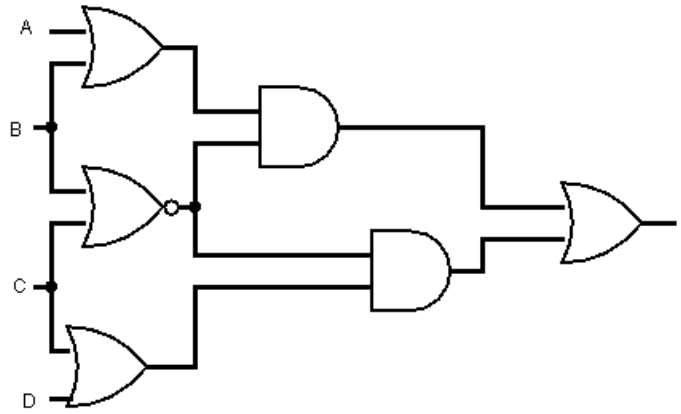
Identify the ordered triples that make the following circuit TRUE:



A. (1, *, 1), (1, 0, 0)
B. (0, 1, 1), (1, 0, 0)
C. (0, *, 1), (1, 0, *)
D. (1, 1, 1), (0, 0, 0)
E. (1, 0, 1), (*, 0, 0)

4. Digital Electronics

Simplify the following to a Boolean expression that uses the fewest number of gates possible:



A. $\overline{B}\,\overline{C}\,(C + D)$
B. $\overline{B}\,\overline{C}\,(A + D)$
C. $\overline{B}\,\overline{C}\,(A + C)$
D. $\overline{B}\,\overline{C}\,(B + D)$
E. $\overline{B}\,\overline{C}\,(A + B)$

5. Assembly Language

What is printed when the following program is executed?

```
N     DC      60          STOP  LOAD    S
      LOAD    N                 ADD     F
TOP   SUB     =2                STORE   S
      BL      DONE              LOAD    P
      LOAD    =1                STORE   N
LOOP  ADD     =1                BU      TOP
      STORE   F           DONE  PRINT   S
      LOAD    N                 END
      DIV     F           S     DC      0
      STORE   P           F     DC      0
      MULT    F           P     DC      0
      SUB     N
      BE      STOP
      LOAD    F
      BU      LOOP
```

A. 4
B. 5
C. 10
D. 12
E. 16